MILITARY SPECIFICATION

PACKING, PREFORMED

1. SCOPE

1.1 This specification covers requirements, quality assurance provisons, and the preparation for delivery criteria for one type of packing, preformed used on XM454 and XM455 projectiles.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-P-116 - Preservation Methods of

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1169 - Packaging, Packing and Marking for Shipments of Inert Ammunition Components

DRAWINGS

ORDNANCE CORPS

8808362 - Packing and Marking for Storage and Shipments
8848472 - Bag, Packing, Ammunition for Gasket
8857484 - Packing, Preformed
8883537 - Bag, Packing for Repair Parts

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EL8857484-2 - Equipment List
EL8857484-3 - Equipment List
EL8857484-4 - Equipment List
EL8857484-5 - Equipment List
EL8857484-6 - Equipment List
EL8857484-7 - Equipment List

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The packing shall comply with all requirements specified on the drawing(dwg) 8857484, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 Workmanship.

3.3.1 Rubber characteristics. All components shall be free from cold flow, shrinkage, cold shuts, inclusions, porosity or any similar characteristics.

3.3.2 Foreign matter. Parts shall be free from chips, dirt, grease or any embedded foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection. Not applicable.

4.3 Quality conformance inspection.

4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing

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revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. Critical characteristics shall require 100 percent inspection of all items in the inspection lot. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table I and III major and Table II and IV minor), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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Table I

Major Characteristics For War Reserve Items

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

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Table II
Minor Characteristics For War Reserve Items

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

Table III
Major Characteristics For Training Items

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

Table IV
Minor Characteristics For Training Items

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-9	all	14-16	12	28-33	16	77-121	20
10	9	17-18	13	34-41	17	122-273	21
11	10	19-22	14	42-54	18	over 273	22
12-13	11	23-27	15	55-76	19		

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4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Packing preformed (see dwg 8857484).

Categories	Characteristics	Method of Inspection
Critical:	None defined	
Major:		
	101. Axial thickness (see 4.4.1)	Gage
	102. Inside diameter (see 4.4.2)	Gage
	103. Properly formed no splits or cracks	Visual
	104. Surface has no defects, blemished or irregularities	Visual
Minor:		
	201. Workmanship	Visual

4.3.2.1.2 Preparation for delivery (see 5.1).

Categories	Characteristics	Method of Inspection
Critical:	None defined	
Major:		
	101. Preservation, packaging, packing, and marking in accordance with MIL-STD-1169	Visual/Manual
Minor:	None defined.	

4.3.2.1.3 Bag, packing for repair parts (see dwg 8883537) and bag, packing, ammunition for gasket (see dwg 8848472).

Preformed packing Dwg Part No	Ord Pkg part No. (see dwgs 8883537 & 8848472)
8857484-2	8883537-6
8857484-3	8883537-6
8857484-4	8848472-3
8857484-5	8883537-6
8857484-6	8848472-4
8857484-7	8848472-4

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Categories	Characteristics	Method of Inspection
Critical:	None defined	
Major:		
	101. Material (see 3.1)	Visual
	102. Length before packing, min	*Measure
	103. Width before packing, min	*Measure
	104. Properly packed	Visual/Manual
	105. Properly heat sealed	Visual/Manual
	106. No split or crack in bag	Visual
	107. Objective evidence (see 6.2) for heat seal test in accordance with MIL-P-116	Visual

*Measure with suitable scale.

Minor:		
	201. Marking	Visual

4.3.2.1.4 Packing and marking for storage and shipments (see dwg 8808362).

Categories	Characteristics	Method of Inspection
Critical:	None defined	
Major:		
	101. Gross load weight (max)	*Measure
	102. Contents tight in package	Manual
	103. Correctly packed	Visual/Manual
	104. Material (see 3.1)	Visual

*Determine weight with suitable scale.

Minor:		
	201. Marking	Visual

4.3.3 Testing. Not applicable

4.3.4 Acceptance inspection equipment.

4.3.4.1 Critical and major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL) (See EL-8857484 , -2, -3, -4, -5, -6, -7 as applicable). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

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a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Axial thickness (see dwg 8857484). A micrometer with 1/4 inch diameter ball points shall be used to determine axial thickness (dimension B). In employing this instrument the tips are closed, and the packing (or section thereof) allowed to rest on the tips, and the tips very carefully separated until the packing falls thru of its own weight. Three measurements shall be taken at points approximately 120 degrees apart around the circumference. The average of the three readings shall comply with the requirements of the applicable detail on dwg 8857484.

4.4.2 Inside diameter (see dwg 8857484). The inside diameter shall be measured with a stepped cone employing no stress in applying the packing in excess of that necessary to overcome the ellipticity of the packing.

5. PREPARATION FOR DELIVERY

5.1 Production items.

5.1.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1.1 Level A. The packing shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.1.1.2 Level C. The packing shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.1.2 Packing.

5.1.2.1 Levels A and C. The packing shall be packed in accordance with the applicable instructions contained in MIL-STD-1169(MU).

5.1.3 Marking. Marking shall be in accordance with MIL-STD-1169(MU).

5.2 Repair parts.

5.2.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

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5.2.1.1 Level A. The packing shall be packaged in accordance with dwg 8883537 or 8848472 as applicable and 8808362 using the specified moisture barrier.

5.2.1.2 Level C. The packing shall be packaged in accordance with dwg 8808362 except that the moisture barrier may be omitted.

5.2.2 Packing.

5.2.2.1 Level A and C. The packing shall be packed in accordance with dwg 8883537 or 8848472 as applicable and 8808362.

5.2.3 Marking. Marking shall be as specified in accordance with dwg 8883537 or 8848472 as applicable and 8808362.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.
b. Production or Repair Parts.
c. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).
d. Training or War Reserve Item

6.1.2 Contract data requirements (see 3.4.3.1). Data specified in 3.4.3.1 will be delivered as identified on a numbered DD Form 1664 or specification when specified on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SMUPA-QA-I.

Custodian:
Army - MU

Preparing activity:
Army - MU

Project No. 1110-A028

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